

Lecture 9: Bayes Theorem

29 June 2026 12:40

Topics of this lecture:

1. Bayes Theorem

According to conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

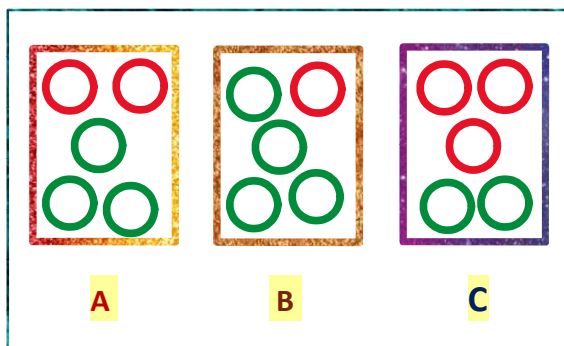
According to multiplication theorem:

$$P(A \cap B) = P(B | A) \cdot P(A)$$

Now we can write:

$$P(A|B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

This is the expression of Bayes Theorem



Q. Pick a bag at random
And draw a random ball out of it.

1. Probability of getting a red ball given bag a is chosen ?

Conditional Probability

$$P(\text{Red Ball} | A) = 2/5$$

2. Probability of getting a red ball and choosing bag A ?

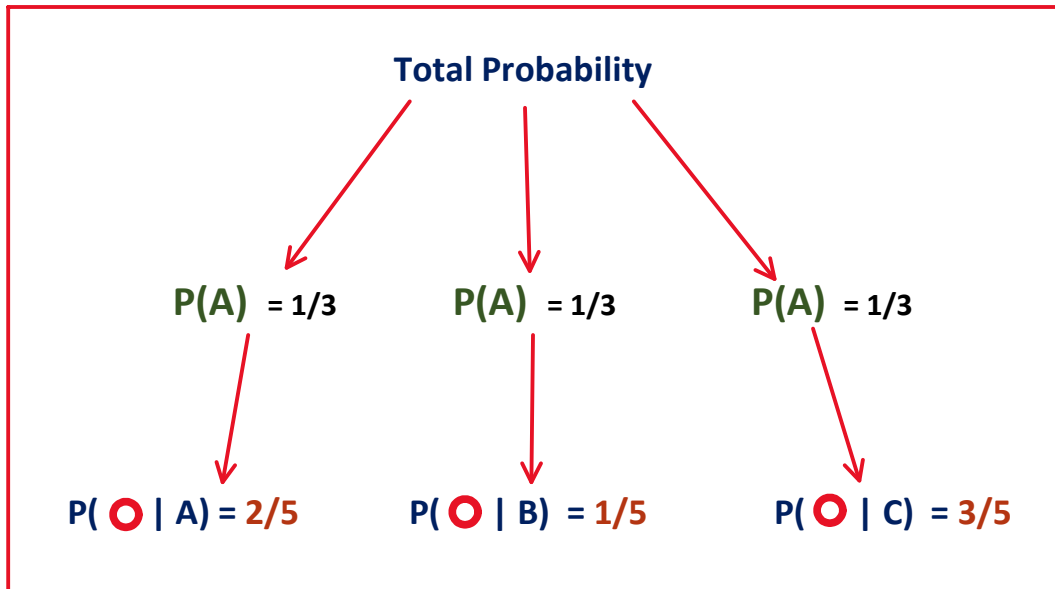
Multiplication Theorem

$$\begin{aligned} P(\text{Red Ball} \cap A) &= P(\text{Red Ball} | A) \cdot P(A) \\ &= 2/5 \cdot 1/3 \\ &= 2/15 \end{aligned}$$

3. Probability of getting a red ball ?

Total Probability

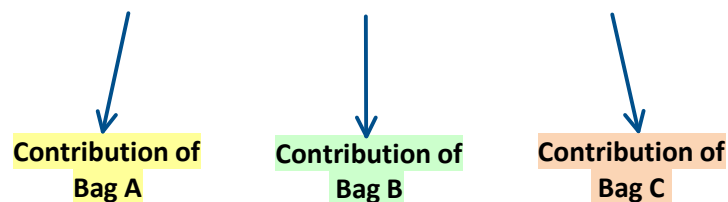
$$\begin{aligned}
 P(\text{Red}) &= P(\text{Red} | A) \cdot P(A) + P(\text{Red} | B) \cdot P(B) + P(\text{Red} | C) \cdot P(C) \\
 &= \frac{2}{5} \cdot \frac{1}{3} + \frac{1}{5} \cdot \frac{1}{3} + \frac{3}{5} \cdot \frac{1}{3} \\
 &= \frac{2}{5}
 \end{aligned}$$



4. Probability of bag a is chosen given that the drawn ball is red ?

Total Probability

$$P(\text{Red}) = P(\text{Red} | A) \cdot P(A) + P(\text{Red} | B) \cdot P(B) + P(\text{Red} | C) \cdot P(C)$$



$$P(A | \text{Red}) = \frac{P(\text{Red} \cap A)}{P(\text{Red})} = \frac{\text{Contribution of Bag A}}{\text{Total Probability of red ball}}$$

$$P(A | \text{Red}) = \frac{\frac{2}{5} \cdot \frac{1}{3}}{\frac{2}{5} \cdot \frac{1}{3} + \frac{1}{5} \cdot \frac{1}{3} + \frac{3}{5} \cdot \frac{1}{3}}$$

$$P(A | \text{Red}) = \frac{\frac{2}{15}}{\frac{2}{5}} = \frac{1}{3}$$

Q2. Suppose two factories supply light bulbs to the market

- Factory A's bulb work for over 5000 hours 99% of the cases.
- Factory B's bulb work for over 5000 hours 95% of the cases.

Factory A supplies 60% of the total bulbs.

What is the chance that a randomly purchased bulb will work for more than 5000 hours ?

Let event E = randomly purchased bulb will work for more than 5000 hours

Total Probability

$$P(E) = P(E | A) \cdot P(A) + P(E | B) \cdot P(B)$$

$$P(E) = 0.99 \cdot 0.6 + 0.95 \cdot 0.4$$

$$P(E) = 0.97$$

Given that -

The purchased bulb worked for more than 5000 hours.

What is the probability that it was produced by factory A?

Bayes Theorem

$$P(A | E) = \frac{\text{Contribution of Factory A}}{\text{Total Probability of E}} = \frac{P(E | A) \cdot P(A)}{P(E | A) \cdot P(A) + P(E | B) \cdot P(B)}$$

$$P(A | E) = (0.99 \times 0.6) / (0.97)$$

$$= 0.6124$$

probability that it was produced by factory A is 61 %